Short Windows, Unmodelled Costs, Unknown Cutoffs: A Pre-Registered Evidence Audit of 91 LLM Trading Papers

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Abstract

Papers on large-language-model (LLM) trading agents and alpha mining report Sharpe ratios that would be remarkable in any other part of empirical finance. We audit the evidence behind those numbers. Before coding a single paper we pre-registered a population of 91 papers from a curated corpus of LLM trading research, a ten-field coding rubric, an analysis script and five directional hypotheses, and committed all of them to version control. Every code is backed by a verbatim quote that a script checks against the paper's text, and a second, blind coder recoded a random 20%. Of the 75 papers with a trading test, 51% deduct no trading costs and only 9% model costs from data; the median test window is 12 months, and a quarter of windows are six months or shorter. Among the 51 papers whose test window is not known to lie after the model's training cutoff, 98% neither test after the cutoff, nor evaluate live, nor use a chronologically consistent model. Only 9% report a multiplicity-adjusted result, and just five papers combine modelled costs, a leakage control and a reported trial count. Two of the five hypotheses pass (short windows; uncontrolled look-ahead). Three fail: more papers report a trial count than we predicted, though most of those counts are repeated random seeds rather than the number of strategies searched; and the predicted links between headline Sharpe and test-window length, and between headline Sharpe and costs, are not detectable in the 13 papers that state a headline Sharpe in their abstract or conclusion. We release the codes, quotes and script, and propose a six-line reporting minimum.

1 Introduction

Empirical finance learned, at some cost, that a backtest is a claim about a search as much as about a market. Data snooping in asset pricing is an old problem [14, 16, 7]; most published return predictors do not survive correction for the number of tests tried [8, 10]; backtest overfitting is easy to produce and hard to detect after the fact [1, 2]; and published predictability decays once it is known [15].

LLM trading research has rebuilt much of this risk in a new form. A language model may have read the test period during pre-training, so a backtest can measure memory rather than reasoning [11, 12, 5]. Systems that generate many candidate strategies select the best one, often without reporting how many were tried [6]. And evaluations are often short and narrow: when timing-based LLM strategies are re-run over two decades and more than a hundred symbols, their reported advantage largely disappears [13].

What has been missing is a count. How many papers in this literature model costs, control look-ahead, or report their search? The closest prior work, a PRISMA-ScR scoping review of execution

realism in LLM trading systems [18], deliberately withholds its field-level counts pending author validation. We provide those counts for a defined population, under a design that makes the audit itself hard to bend. The population, rubric, hypotheses and analysis script were committed before any paper was coded; every code carries a quote that a script checks against the source text; a blind second coder recodes a random fifth; and every change after coding is logged.

Our results are mixed in the way a pre-registered study should allow. The evidence base is thin on costs and look-ahead control: half of trading tests deduct no costs, and among papers whose test may overlap the model's training data, almost none control for it. But two of the strongest-sounding hypotheses fail: trial counts are reported more often than we expected, and there are too few headline Sharpe ratios in abstracts to detect the associations we predicted.

2 Design

Population. The corpus is a curated collection of 233 papers on LLM trading, text-based signals, benchmarks and the classic premia and replication literature, each with its full text extracted. The population is every paper in themes 00–07 that is an LLM agent or alpha-mining paper, or that reports a headline Sharpe ratio or annual return, and whose full text is available: 91 papers. By kind, they are 50 agent papers, 13 benchmarks, 11 alpha-mining papers, 8 other methods, 5 measurement papers and 4 surveys. Most are 2024–2026 preprints. The list was frozen before coding.

Rubric. Each paper's main reported trading experiment (the one its abstract describes) is coded on ten fields. *test_months* is the length of the out-of-sample or test window, and *n_assets* is the number of tradable assets. *costs* is none, a flat stated rate, or realistic (spread, impact or borrow modelled from data). *leakage* is none, acknowledged only, a test after the model's training cutoff, a live or forward evaluation, or a chronologically consistent model [9]. *trials* is none, mentioned, a count reported, or a multiplicity-adjusted result (deflated Sharpe, reality check, false-discovery control). The rubric also records a buy-and-hold baseline, the headline Sharpe as stated in the abstract or conclusion, public code, and whether the main model's training cutoff is known to precede the test window.

Hypotheses. The directions were fixed in advance:

H1 Fewer than 10% of papers report how many strategies or configurations were tried, and fewer than 5% report a multiplicity-adjusted result.

H2 The median test window is 12 months or less.

H3 Headline Sharpe is negatively rank-correlated with test-window length (one-sided permutation test, $p < 0.05$).

H4 Among papers whose test window is not known to lie wholly after the model's training cutoff, a majority do not address training leakage.

H5 Papers that model costs report lower headline Sharpe ratios than those that do not (one-sided permutation Mann–Whitney test, $p < 0.05$).

Coding and checks. Coding was done by language-model coders working from the full text under the fixed rubric. Each non-empty code had to carry a 5–30-word verbatim quote. A script collapses whitespace and checks every quote as a substring of the paper's extracted text, and none failed. A blind second coder recoded 18 randomly chosen papers (seed fixed in advance). The two

Table 1: Evidence fields across 75 papers with a trading test.

Field and value	Papers	Share
Trading costs		
none deducted	38	51%
flat stated rate	30	40%
modelled from data	7	9%
Look-ahead and training leakage		
not discussed	30	40%
acknowledged only	24	32%
test after training cutoff	14	19%
live or forward	6	8%
chronologically consistent model	1	1%
Search and multiplicity		
not discussed	39	52%
mentioned	6	8%
count reported	23	31%
adjusted result	7	9%
Other		
buy-and-hold or index baseline	47	63%
public code	22	29%
costs, leakage control and trial count together	5	7%

coders agree on costs in 100% of papers, on the buy-and-hold baseline in 94%, on leakage, trials and code in 89%, on test-window length (within 10%) in 94%, and on the headline Sharpe (within 5%) in 94%. That LLMs did the coding is itself a risk, and the quote check and second pass are there to bound it; Section 5 discusses what they do not catch.

Changes after coding began. Three, all logged. Before any result was computed, two codes were made consistent with the rubric: one paper's live phase was coded instead of its main backtest, and one five-day window was coded as zero months. The first run's H4 code selected papers by their leakage code rather than by the pre-registered cutoff criterion, which made the share 100% by construction. We fixed the script to match the pre-registration; no codes changed, and the first run's output is kept.

3 Results

Sixteen of the 91 papers have no trading test, being benchmarks of reasoning, forecasting accuracy, conceptual papers or reviews. The results below are for the remaining 75 unless stated otherwise.

Costs (Table 1). Half the trading tests deduct no costs at all. Forty per cent apply a flat rate, typically 5–20 basis points or a fixed commission, and only seven papers (9%) model spread, impact or borrow from data.

H2 passes: windows are short. Sixty-nine papers state a test window. The median is 12 months, with quartiles of 6 and 36 months, and 18 papers test on six months or less. The median

test uses 69 assets, but 20 of 54 papers that state a count test on ten assets or fewer, often single stocks traded one at a time.

H4 passes: look-ahead is rarely controlled. For 24 papers the main model's training cutoff is known to precede the test window, for 22 it is known not to, and for 29 it cannot be determined from the paper. Of the 51 papers in the last two groups, 50 (98%) neither test after the cutoff, nor evaluate live, nor use a chronologically consistent model: their test may overlap what the model read during training, and they either say nothing about it (as in 40% of all trading tests) or acknowledge it without control. Live evaluation, the cleanest control [11, 4], appears in six papers.

H1 fails, with a qualification. Thirty papers (40%) report a trial count or an adjusted result, four times the 10% we predicted, and seven (9%) report an adjusted result against a predicted 5%. The qualification is in what is counted. Reading the 23 quoted counts, 13 are repeated runs of one configuration over random seeds (typically three to five), which measures run-to-run variance, not search; ten count candidates or configurations searched, the quantity a deflated Sharpe ratio needs. On the narrower reading, 17 papers (23%) disclose their search or adjust for it. The hypothesis still fails at that reading, and we report it as failed. This split is exploratory; the rubric did not separate the two cases.

H3 and H5 fail: too few headline Sharpe ratios. Only 15 papers state a Sharpe ratio for their method in the abstract or conclusion, and 13 after excluding relative improvements. Their median is 2.0. Within those 13, the rank correlation between headline Sharpe and window length is -0.10 (one-sided permutation $p = 0.36$). The cost comparison has 11 papers that model costs and two that do not; the medians are 1.62 and 3.46, in the predicted direction, with one-sided $p = 0.10$. Neither test can detect the predicted association at this sample size, and we do not read the cost medians as evidence. That most papers keep their Sharpe ratio out of the abstract is itself a finding about reporting. Many quote one in the introduction or a table, often per asset or as gross excess.

Robustness. One paper appears twice in the population, as a preprint and as its published version. Removing the duplicate leaves every verdict unchanged (H3: $\rho = -0.23$, $p = 0.24$; H5: $p = 0.10$).

4 What the counts mean

The pattern is consistent with a field that has adopted the vocabulary of careful evaluation faster than its practice. A buy-and-hold baseline is standard (63%), and repeated seeds are common. The controls that address the two failure modes particular to LLM backtests are rare: costs that would survive contact with execution, and protection against a model that has read the test period. Only five papers combine modelled costs, a leakage control and a reported trial count, and one of those also releases code.

These are not reasons to dismiss individual papers. Several in the population are designed around exactly these controls: live benchmarks [11], forward-logged nowcasts [4], leakage-safe discovery with trial-count deflation [6], production records [3], audits that separate an agent's selection skill from passive exposure [17], and contamination tests [5]. The counts say these are exceptions.

A reporting minimum. Six lines would let a reader judge any LLM trading result in a minute:

1. the test window, dated, and the number of assets;
2. the main model and its training cutoff, and whether the test lies after it;
3. the cost model, with the breakeven cost at which the result vanishes;
4. the number of strategies, prompts or configurations searched before the reported one;
5. a multiplicity-aware statistic (deflated Sharpe or a reality check) at that count;
6. a benchmark-relative or composition-matched null [17].

A research loop can produce these lines by construction rather than by discipline. Our companion paper [20] describes one that records every trial, pre-registers each hypothesis and spends its holdout once, in an open engine [19].

5 Limitations

The population is one curated corpus, not a systematic search, so the shares describe that corpus. They are likely representative of widely circulated work, since the corpus collects cited and recent papers, but that is not guaranteed. The coders were language models. The quote check guarantees that each code points to real text, not that the judgement is right; the second pass bounds disagreement at about one paper in ten on the subjective fields, on 18 papers only. Where a paper gives no cutoff, we code it unknown rather than look up the model's published cutoff, which makes H4 conservative in one direction and leaves 29 papers undetermined. The headline-Sharpe field was defined strictly (abstract or conclusion), which kept the H3 and H5 samples too small for their tests. A broader definition was not pre-registered, so we do not report one. Finally, the author builds and sells research tooling in this area; the codes, quotes and script are released so the counts can be checked without trusting us.

6 Conclusion

Half the trading tests in this population deduct no costs, the median window is a year, and almost no paper whose test may overlap its model's training data controls for that overlap. Search counts are reported more often than we expected, but mostly as repeated seeds rather than as the size of the search. None of this shows that LLM trading strategies fail. It shows that most of their published evidence could not tell us if they did.

Data and code. The pre-registration (committed before coding), the rubric, the 91 coded records with their quotes, the blind second pass, the correction log and the analysis script are released at `https://github.com/henryzhangpku/llm-trading-evidence-audit`.

About the author. Henry Zhang is a quantitative engineer with fifteen years building research platforms and production trading systems, including eight years at BlackRock and two at PIMCO. The research engine used here is open source at `https://github.com/henryzhangpku/autonomous-quant-researcher`; more research and notes at `https://henryzhang.substack.com` and `https://www.officialhenryzhang.com`.

Disclaimer. This is a study of published research. It is not investment advice.

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